

$$\begin{aligned} &> f(b, c, r) := \sin^2(b) + \sin^2(c) + 2 \cdot \sin(b) \cdot \sin(c) \cdot \cos(b + c + r) \\ &\quad f := (b, c, r) \mapsto \sin(b)^2 + \sin(c)^2 + 2 \cdot \sin(b) \cdot \sin(c) \cdot \cos(b + c + r) \end{aligned} \quad (1)$$

$$\begin{aligned} &> g(b, c, r) := \text{simplify}(\text{expand}(f(b, c, 0)^2 - f(b, c, r) \cdot f(b, c, -r))) \\ &\quad g := (b, c, r) \mapsto \text{simplify}(\text{expand}(f(b, c, 0)^2 - f(b, c, r) \cdot f(b, c, -r))) \end{aligned} \quad (2)$$

$$\begin{aligned} &> \text{collect}(\text{expand}(g(b, c, r)), \cos(r)) \\ &-4 \sin(c)^2 \sin(b)^2 \cos(r)^2 + (4 \cos(b)^3 \cos(c) \sin(c) \sin(b) - 4 \cos(b)^2 \sin(c)^2 \sin(b)^2 \\ &\quad + 4 \cos(b) \cos(c)^3 \sin(c) \sin(b) - 4 \cos(c)^2 \sin(c)^2 \sin(b)^2 - 8 \cos(b) \cos(c) \sin(c) \sin(b) \\ &\quad + 8 \sin(c)^2 \sin(b)^2) \cos(r) - 4 \cos(b)^3 \cos(c) \sin(c) \sin(b) + 4 \cos(b)^2 \sin(c)^2 \sin(b)^2 \\ &\quad - 4 \cos(b) \cos(c)^3 \sin(c) \sin(b) - 4 \sin(c)^4 \sin(b)^2 + 8 \cos(b) \cos(c) \sin(c) \sin(b) \end{aligned} \quad (3)$$

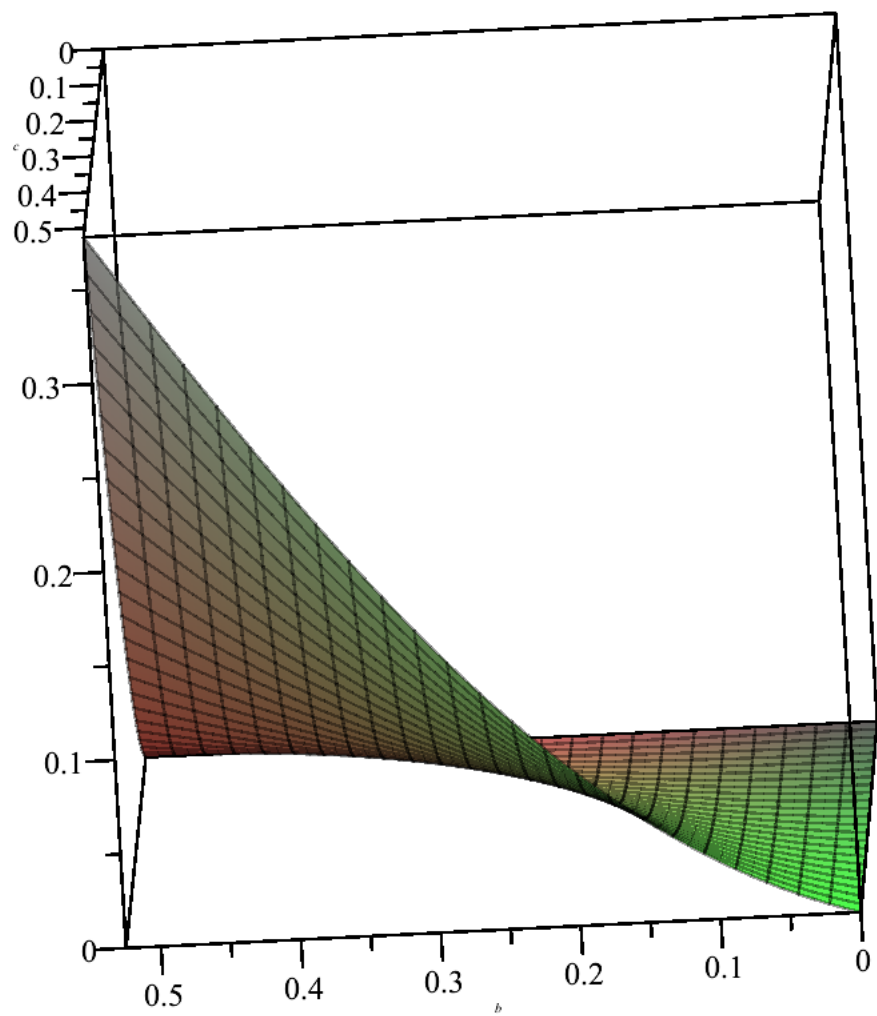
$$\begin{aligned} &> \text{simplify}\left(\frac{1}{r^2} \left(-4 \sin(b)^2 \sin(c)^2 \cdot (1 - r^2) + (-4 \sin(b)^2 \sin(c)^2 \cos(c)^2 \right. \right. \\ &\quad - 4 \sin(b)^2 \sin(c)^2 \cos(b)^2 + 4 \sin(b) \sin(c) \cos(c)^3 \cos(b) \\ &\quad + 4 \sin(b) \sin(c) \cos(c) \cos(b)^3 + 8 \sin(b)^2 \sin(c)^2 - 8 \sin(b) \sin(c) \cos(b) \cos(c) \left. \right) \cdot \left(1 \right. \\ &\quad \left. - \frac{r^2}{2} \right) - 4 \sin(c)^4 \sin(b)^2 + 4 \sin(b)^2 \sin(c)^2 \cos(b)^2 - 4 \sin(b) \sin(c) \cos(c)^3 \cos(b) \\ &\quad \left. - 4 \sin(b) \sin(c) \cos(c) \cos(b)^3 + 8 \sin(b) \sin(c) \cos(b) \cos(c) \right) \Bigg) \\ &-2 \sin(b) (\cos(c) \cos(b)^3 - \cos(b)^2 \sin(c) \sin(b) + (\cos(c)^3 - 2 \cos(c)) \cos(b) \\ &\quad - \cos(c)^2 \sin(c) \sin(b)) \sin(c) \end{aligned} \quad (4)$$

$$\begin{aligned} &> h(b, c) := -2 (\cos(c) \cos(b)^3 - \sin(b) \sin(c) \cos(b)^2 + (\cos(c)^3 - 2 \cos(c)) \cos(b) \\ &\quad - \sin(b) \sin(c) \cos(c)^2) \sin(b) \sin(c) \\ &h := (b, c) \mapsto - (2 \cdot \cos(c) \cdot \cos(b)^3 - 2 \cdot \sin(b) \cdot \sin(c) \cdot \cos(b)^2 + 2 \cdot (\cos(c)^3 - 2 \cdot \cos(c)) \cdot \cos(b) \\ &\quad - 2 \cdot \sin(b) \cdot \sin(c) \cdot \cos(c)^2) \cdot \sin(b) \cdot \sin(c) \end{aligned} \quad (5)$$

$$\begin{aligned} &> \text{simplify}(h(b, c) - 2 \cdot (\sin^2(b) + \sin^2(c)) \cdot \cos(b + c) \cdot \sin(b) \sin(c)) \\ &\quad 4 \sin(c)^2 \sin(b)^2 \end{aligned} \quad (6)$$

$$\begin{aligned} &> \text{simplify}\left(\text{int}\left(\text{int}\left(\frac{h(b, c)}{\sin(b + c)^4}, b = 0 \dots \frac{\pi}{6}\right), c = 0 \dots \frac{\pi}{6}\right)\right) \\ &\quad -\frac{1}{4} + \frac{\pi\sqrt{3}}{24} + \frac{\ln(3)}{8} \end{aligned} \quad (7)$$

$$> \text{plot3d}\left(h(b, c), b = 0 \dots \frac{3.15}{6}, c = 0 \dots \frac{3.15}{6}\right)$$



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